

Aadarsha Gopala Reddy

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SUMMARY

MS Computer Science graduate from Washington University in St. Louis with hands-on experience in machine learning and data analytics. Skilled in building end-to-end ML pipelines, data-intensive applications, and distributed systems using Python, TensorFlow, SQL, Spark, and cloud technologies. Seeking software, data, or ML engineering roles.

SKILLS

Programming Languages: Java, C++, Python, R, HTML, CSS, JavaScript/TypeScript, PHP, C#, SQL, Rust.

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Node.js, Vue.js, React, Socket.IO, Gemini/OpenAI API.

Tools & Technologies: Git, Tableau, Power BI, MySQL, AWS, MongoDB, Microsoft 365, Google Workspace.

Core Competencies: Analytical Skills, Problem Solving, Critical Thinking, Communication, Team Collaboration, Leadership, Adaptability, Project Management.

Languages: Proficient in English, Kannada, Telugu; Conversational in Hindi.

EDUCATION

M.S. in Computer Science

August 2024 - May 2026

Washington University in St. Louis, St. Louis, Missouri, USA

- **GPA:** 3.67/4.00
- *Honors:* Bauer Leaders Academy Leadership Development Badge.
- *Coursework:* Computer Vision, Data Manipulation & Management at Scale, Deep Learning, Systems Security, Software Development, and Quantum Computing.

B.A. in Computer Science and Data Analytics; Economics Minor

August 2020 - May 2023

Ohio Wesleyan University, Delaware, Ohio, USA

- **GPA:** 3.42/4.0
- *Honors:* Golden Bishop Award, Mortar Board, Florence Leas Prize, Spring 2022 and Spring 2023 Dean's List.
- *Coursework:* Computer Architecture, Theory of Computation, Algorithms, Big Data, Data Visualization, Data Analytics, Databases, Machine Learning, Artificial Intelligence, Applied Statistics.

EXPERIENCE

Graduate Teaching Assistant - CSE 5114: Data Manipulation and Management at Scale

January 2026 - May 2026

Washington University in St. Louis, St. Louis, Missouri, USA

- Guided students in building data pipelines and real-time processing systems using Python, Snowflake, SQL, Airflow, Spark, Kafka, and Flink.
- Conducted oral midterm exams simulating technical interviews on distributed storage, fault tolerance, and scaling; mentored capstone project groups on data architecture.

Graduate Assistant with the Taylor Family Center for Student Success

November 2025 - May 2026

Washington University in St. Louis, St. Louis, Missouri, USA

- Led data analysis and storytelling initiatives, tracking program impact through SA Tools, survey design, and social media analytics for evidence-based decision-making.
- Administered the TFCSS website and developed process manuals to establish best practices in data evaluation for staff.

AI Engineer Intern

June 2025 - August 2025

Crittero, Inc., Remote

- Engineered a full-stack social media simulation and recommendation system with persona-based data generation in TypeScript and a recommendation algorithm using TensorFlow/Keras.
- Built a Python testing infrastructure for ML performance validation, working across the end-to-end ML lifecycle from data generation to model training.

Graduate Assistant with the Taylor Family Center for Student Success Washington University in St. Louis, St. Louis, Missouri, USA Led FLI student success initiatives and pioneered data assessment of social media and newsletters to translate user interaction insights into strategies that measurably improved engagement.	August 2024 - May 2025
Data Analyst Intern Lab714, Boca Raton, Florida, USA - Achieved a 20% reduction in costs for clients through data-driven insights. - Engineered a software using React and AWS to extract, organize, and analyze data from IoT devices.	June 2023 - May 2024

RESEARCH EXPERIENCE

Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data DRIVES Project, Washington University in St. Louis, St. Louis, Missouri, USA - Built an end-to-end pipeline processing 26,968 participant-weeks of naturalistic driving data from 304 participants, deriving weekly driving features and integrating demographic covariates for CDR status prediction. - Compared six CDR prediction models (GRU-DANN, DANN, logistic regression, random forest, XGBoost, MLP) under leave-one-group-out cross-validation; GRU-DANN achieved highest participant-level ROC AUC (0.599) and balanced accuracy (0.584).	August 2025 - May 2026
AI in Modern Board Games (Lost Cities) Ohio Wesleyan University, Delaware, Ohio, USA Built a Java implementation of Lost Cities and an AI agent; achieved 13/18 wins against a human baseline in testing.	May 2022 - July 2022

LEADERSHIP

Graduate and Professional Student Council (GPSC) Washington University in St. Louis <i>Vice President of the Graduate-Professional Council (GPC) Chamber</i> - Managed the Council’s unified Microsoft Teams workspace and website, establishing centralized communication hubs and maintaining up-to-date information on events, governance, and resources. - Represented the graduate student body to university leadership and supported the GPC President in advancing strategic council priorities. - Chaired the Constitution Committee (Aug 2025 - Feb 2026), coordinating stakeholder input to draft and ratify GPSC governing documents.	May 2025 - May 2026
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PROJECTS

Datacenter Cooling Optimization using Deep Reinforcement Learning - Implemented multiple DRL algorithms (DDQN, PPO, SAC) integrated with EnergyPlus simulations for datacenter cooling optimization, achieving up to 35.8% improvement over baseline controllers. - Developed custom environment using Sinergym that simulates a small datacenter with stochastic weather conditions, focusing on optimizing energy efficiency in small to mid-sized facilities.	August 2024 - December 2024
Interactive Storybook - Built an interactive app with Vue.js, Node.js, and MongoDB, enabling AI-driven content generation and branching story narratives. - Integrated OpenAI API for dynamic text/image generation, offering multiple story progression choices.	August 2024 - December 2024

REFERENCES

Available upon request.